

Systems of exponential and logarithmic equations

Two equations, one substitution — and the domain that has to survive both.

LESSON 63–64

2 × 40 MIN

ALGEBRA 10, §3.7

EXTENSION

LESSON CLOCK



What is new	10 min
Exponential systems	24 min
Logarithmic systems	24 min
Mixed systems	22 min
Practice	20 min
Homework	10 min

LEARNING OBJECTIVES

- Solve a system by substituting $u = a^x$ and $v = a^y$.
- Solve a system by substituting $u = \log x$ and $v = \log y$.
- Track the domain conditions of both equations.
- Check every candidate pair in the original system.

TEXTBOOK

National	pp. 261–272
Cambridge	Pure Mathematics 2 & 3, pp. 39–44
Workbook	P2 Exercise 2F

01 Explanation

What is new, and what is not

Nothing about the algebra is new: the systems are solved by substitution and elimination exactly as in Quarter II. What is new is the bookkeeping — two domains, two substitutions and two sign conditions to keep straight.

TWO SUBSTITUTIONS, TWO DIFFERENT CONDITIONS

$u = a^x$ carries $u > 0$. $u = \log x$ carries **no** condition on u , but the original x must satisfy $x > 0$. Write the right one down.

Exponential systems

$$2^x + 3^y = 17 \quad 2^{x+2} - 3^{y+1} = 5$$

Put $u = 2^x > 0$ and $v = 3^y > 0$. The second equation is $4u - 3v = 5$, so with $u + v = 17$:

$$4u - 3(17 - u) = 5 \Rightarrow 7u = 56 \Rightarrow u = 8, v = 9$$

Both positive, so $2^x = 8$ gives $x = 3$ and $3^y = 9$ gives $y = 2$.

EXPAND THE INDICES FIRST

$2^{x+2} = 4 \cdot 2^x$, not $2^x + 4$. Every index law is applied before the substitution, not after.

Logarithmic systems

$$\lg x + \lg y = 3 \lg x - \lg y = 1$$

Domain $x > 0, y > 0$. Adding gives $2 \lg x = 4$, so $\lg x = 2$ and $x = 100$; then $\lg y = 1$ and $y = 10$. Both positive ✓.

An equivalent route condenses first: $\lg(xy) = 3$ gives $xy = 1000$, and $\lg \frac{x}{y} = 1$ gives

$\frac{x}{y} = 10$ — the sum-and-quotient system of Quarter II.

Mixed systems

When one equation is exponential and the other logarithmic, solve the simpler one for a variable and substitute:

$$\log_2 x + y = 5 \quad 2^y = x$$

The second gives $y = \log_2 x$, so the first is $2 \log_2 x = 5$ and $\log_2 x = 2.5$, giving $x = 2^{2.5} = 4\sqrt{2} \approx 5.66$ and $y = 2.5$.

LOOK FOR THE LINK

The two equations almost always share a structure — the same base, or one is the other rearranged. Finding that link is the step; everything after it is Grade 9 algebra.

02 Worked examples

EXAMPLE 1 Solve $2^x + 3^y = 17$, $2^{x+2} - 3^{y+1} = 5$.

① $u = 2^x > 0, v = 3^y > 0$
 $u + v = 17, 4u - 3v = 5$

② $4u - 51 + 3u = 5 \Rightarrow u = 8$

③ $v = 9$

④ $x = 3, y = 2$

ANSWER

$(3, 2)$

EXAMPLE 2 Solve $\lg x + \lg y = 3$, $\lg x - \lg y = 1$.

① Domain $x, y > 0$.

② Add: $2 \lg x = 4 \Rightarrow \lg x = 2$.

③ $x = 100$

④ $\lg y = 1 \Rightarrow y = 10$

ANSWER

$(100, 10)$

EXAMPLE 3 Solve $x + y = 10$, $\lg x + \lg y = 1$.

① Domain $x, y > 0$.

② $\lg(xy) = 1 \Rightarrow xy = 10$

③ $t^2 - 10t + 10 = 0$

④ $t = 5 \pm \sqrt{15}$
Both positive ✓

ANSWER

$(5 + \sqrt{15}, 5 - \sqrt{15})$ and the reverse

03 Interactive model

Write both domains on the board before either substitution.

Which substitution?

For $2^x + 3^y = 17$ put:

The condition on that u is:

$2^{(x+2)}$ equals:

For $\lg x + \lg y = 3$ the condition is:

$$x, y > 0$$

$$x, y \geq 0$$

$$x + y > 0$$

none

$\lg x + \lg y = 3$ means:

$$x + y = 1000$$

$$xy = 1000$$

$$xy = 3$$

$$x + y = 3$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Auxiliary variables	Yordamchi o'zgaruvchilar	Вспомогательные переменные
Combined domain	Umumiy aniqlanish sohasi	Общая область определения
Back-substitution	Teskari almashtirish	Обратная подстановка
Symmetric system	Simmetrik sistema	Симметричная система
Elimination	Yo'qotish	Исключение
Rejected pair	Rad etilgan juftlik	Отброшенная пара
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$u = a^x$ carries:

$u = \lg x$ carries:

$3^{(y+1)}$ equals:

$\lg x - \lg y = 1$ means:

$$x - y = 10$$

$$\frac{x}{y} = 10$$

$$\frac{x}{y} = 1$$

$$x - y = 1$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $2^x = 8, y = x + 1$

ANSWER $(3, 4)$

2. Solve $\lg x = 2, y = 2x$

ANSWER $(100, 200)$

3. Write 2^{x+3} in terms of 2^x

ANSWER $8 \cdot 2^x$

4. Write 3^{y-1} in terms of 3^y

ANSWER $\frac{3^y}{3}$

5. $\lg x + \lg y$ condensed

ANSWER $\lg(xy)$

6. $\lg x - \lg y$ condensed

ANSWER $\lg \frac{x}{y}$

7. Condition for $u = 5^x$

ANSWER $u > 0$

Medium · the standard question

7 PROBLEMS

1. Solve $2^x + 3^y = 17, 2^{x+2} - 3^{y+1} = 5$

ANSWER $(3, 2)$

2. Solve $\lg x + \lg y = 3, \lg x - \lg y = 1$

ANSWER $(100, 10)$

3. Solve $x + y = 10, \lg x + \lg y = 1$

ANSWER $(5 \pm \sqrt{15})$ pairs

4. Solve $2^x \cdot 3^y = 12, 2^x + 3^y = 7$

ANSWER $(2, 1)$ or $(\log_2 3, \log_3 4)$

5. Solve $\log_2 x + y = 5, 2^y = x$

ANSWER $x = 4\sqrt{2}, y = 2.5$

6. Solve $3^x = 9^y, x + y = 3$

ANSWER $(2, 1)$

7. Solve $\lg x = 2 \lg y, x + y = 12$

ANSWER $(9, 3)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $4^x + 4^y = 20, x + y = 3$

ANSWER (1, 2) and (2, 1)

2. Solve $\lg(x + y) = 1, \lg x - \lg y = \lg 4$

ANSWER (8, 2)

3. Solve $2^x + 2^y = 6, 2^{x+y} = 8$

ANSWER (1, 2) and (2, 1)

4. Solve $\log_2 x + \log_2 y = 5, x - y = 6$

ANSWER (8, 4)

5. Solve $x^y = 8, \log_2 x \cdot y = 3$

ANSWER Any x, y with $y \log_2 x = 3$

6. Solve $9^x + 9^y = 90, x + y = 2$

ANSWER (0, 2) and (2, 0)

7. Explain why a pair with $u = -2$ is rejected

ANSWER a^x is never negative

07 Homework

Homework — 5 tasks

Both domains on the first line; both ordered pairs at the end where the system is symmetric.

1. Solve $3^x + 2^y = 17, 3^{x+1} - 2^{y+2} = 7$.
2. Solve $\lg x + \lg y = 4, \lg x - \lg y = 2$.
3. Solve $x + y = 20, \lg x + \lg y = 2$.
4. Solve $2^x = 4^y, x + y = 6$.
5. Explain in three sentences why $u = a^x$ needs a sign condition but $u = \lg x$ does not.